

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL1- CASE STUDY

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Code of Conduct:

I **N.Sai Sankar(192311187)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N. Sai Sankar

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: SoilDry $\rightarrow$ IrrigationNeeded
- 2) P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: SoilDry = True, CropWheat = True

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understan ding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well- structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Case Study 1 — Smart Medical Diagnosis System

(a) Propositional Logic Representation

Symbol definitions:

- F = Patient has Fever
- C = Patient has Cough
- R = Patient has Rash
- B = Patient has Breathlessness
- Fl = Possible Flu
- M = Possible Measles
- Pn = Possible Pneumonia

Rules (knowledge base):

R1: $F \wedge C \rightarrow Fl$

R2: $F \wedge R \rightarrow M$

R3: $C \wedge B \rightarrow Pn$

Facts about Patient A:

F = True, C = True, B = True (R is not asserted, so R = False/unknown)

(b) Forward Chaining — Deriving Diagnoses for Patient A

Forward chaining starts from known facts and fires any rule whose premises are fully satisfied, repeating until no new facts can be derived.

Step	Facts Known So Far	Rule Checked	Fires?	Fact Derived
1	F=T, C=T, B=T	—	—	Initial facts asserted
2	F=T, C=T, B=T	R1: $F \wedge C \rightarrow Fl$	Yes (F=T, C=T)	Fl = True (Possible Flu)
3	F=T, C=T, B=T, Fl=T	R2: $F \wedge R \rightarrow M$	No (R is unknown/False)	No new fact
4	F=T, C=T, B=T, Fl=T	R3: $C \wedge B \rightarrow Pn$	Yes (C=T, B=T)	Pn = True (Possible Pneumonia)
5	F=T, C=T, B=T, Fl=T, Pn=T	All rules re-checked	No further rule fires	Fixpoint reached — stop

- Flu=True
- Pneumonia=True

Measles cannot be concluded because Rash is not known.

(c) Backward Chaining for Pneumonia

The goal is:

Goal=Pneumonia

Backward chaining starts with the goal and asks:

What rule can produce Pneumonia?

Rule 3 produces Pneumonia:

$C \wedge B \rightarrow P$

Therefore, the goal is reduced into two subgoals:

P

↓

$C \wedge B$

Now check the facts:

$C = \text{True}$ $B = \text{True}$

Both subgoals are satisfied.

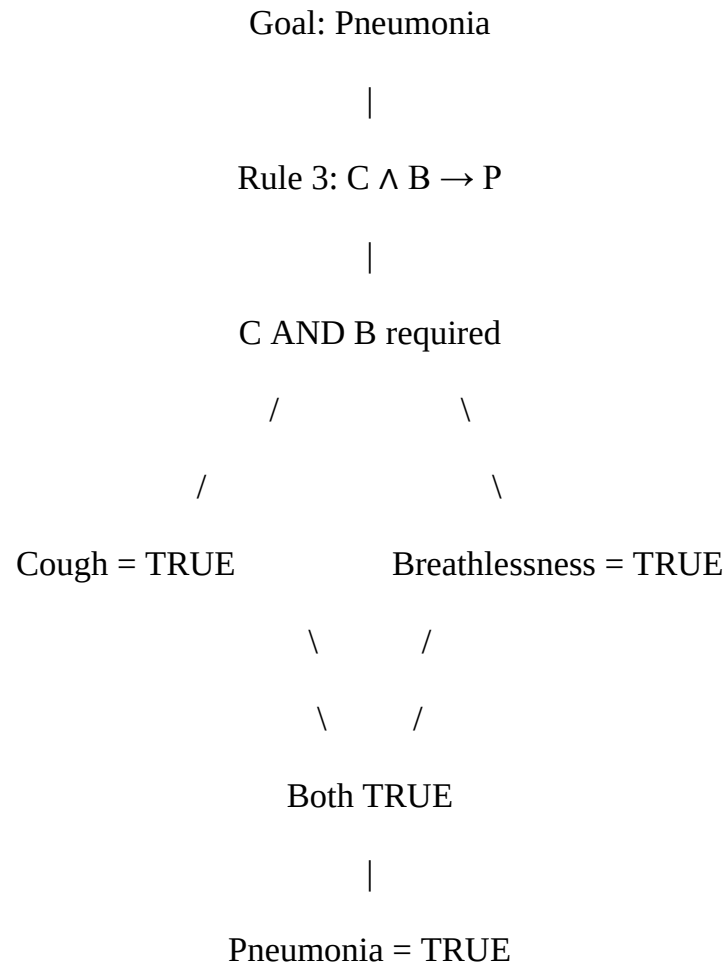
Therefore:

$C \wedge B = \text{True}$

and hence:

Pneumonia=True

Goal Reduction Tree:



Conclusion

The backward-chaining process successfully proves:

Patient A has Pneumonia

This agrees with the forward-chaining result.

Case Study 2 — Autonomous Traffic Management Agent

The FOL knowledge base contains rules for emergency vehicles, clear paths, green signals, and vehicle movement. The given facts include an ambulance and CarA being behind the ambulance.

Predicates

Let:

- $\text{Vehicle}(x)$: x is a vehicle
- $\text{Emergency Type}(x)$: x is an emergency vehicle
- $\text{Clear Path}(x)$: the path for x is clear
- $\text{Green Signal}(x)$: x has a green signal
- $\text{Red Signal}(x)$: x has a red signal
- $\text{Behind}(x, y)$: x is behind y
- $\text{Proceed}(x)$: x can proceed

Given Rules

Rule 1

$x[\text{Vehicle}(x) \wedge \text{Emergency Type}(x) \rightarrow \text{Clear Path}(x)]$

Rule 2

$x[\text{Clear Path}(x) \rightarrow \text{Green Signal}(x) \wedge \neg \text{Red Signal}(x)]$

Rule 3

$x, y[\text{Vehicle}(x) \wedge \text{Behind}(x, y) \wedge \text{Green Signal}(y) \rightarrow \text{Proceed}(x)]$

Facts

$\text{Vehicle}(\text{Ambulance})$ $\text{Emergency Type}(\text{Ambulance})$ $\text{Vehicle}(\text{Car A})$ $\text{Behind}(\text{Car A}, \text{Ambulance})$

Step	Unify	Substitution Applied	Result
1	$\text{Vehicle}(x)$ with $\text{Vehicle}(\text{Ambulance})$	$\theta_1 = \{x/\text{Ambulance}\}$	x is bound to Ambulance
2	$\text{Emergency Type}(x)\theta_1 = \text{Emergency Type}(\text{Ambulance})$ with fact $\text{Emergency Type}(\text{Ambulance})$	$\theta_2 = \{\}$ (already matches)	Consistent — no conflict
3	Combine θ_1 and θ_2	MGU $\theta = \{x/\text{Ambulance}\}$	Unification succeeds

Applying the substitution to Rule 1:

$\text{Vehicle}(\text{Ambulance}) \wedge \text{Emergency Type}(\text{Ambulance}) \rightarrow \text{Clear Path}(\text{Ambulance})$

Since both conditions are true:

$\text{Clear Path}(\text{Ambulance})$

(b) Resolution Proof that Proceed(CarA) is True

Step 1 — Convert all rules to CNF:

R1(CNF): $\neg \text{Vehicle}(x) \vee \neg \text{Emergency Type}(x) \vee \text{Clear Path}(x)$

R2(CNF): $\neg \text{Clear Path}(y) \vee \text{Green Signal}(y)$

R3(CNF): $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{Green Signal}(y) \vee \text{Proceed}(x)$

Facts (unit clauses):

Vehicle(Ambulance), Emergency Type(Ambulance), Vehicle(CarA), Behind (CarA, Ambulance)

Negated goal:

$\neg \text{Proceed}(\text{CarA})$

Resolution Proof:

Vehicle(Ambulance)
+
Emergency Type(Ambulance)

↓
Clear Path(Ambulance)
|
Rule 2
Green Signal(Ambulance)
|
+
|
Vehicle(CarA)
Behind(CarA, Ambulance)
↓
Rule 3 → Proceed(CarA)
|
+
|
¬Proceed(CarA)
|
↓
Contradiction

(c) Two Simultaneous Emergency Vehicles

The current knowledge base can handle multiple emergency vehicles individually, because Rule 1 is universally quantified:

Forall x

However, there is an important limitation.

Suppose there are:

Ambulance1

and

FireTruck1

and both satisfy:

Vehicle(x)

and

Emergency Type(x)

Rule 1 can derive:

Clear Path(Ambulance1)

and:

Clear Path(FireTruck1)

Then Rule 2 can derive green signals for both.

However, the current KB does not explicitly specify priority or conflict resolution when two emergency vehicles require the same road/intersection.

Additional facts needed

For example:

Emergency Priority (Ambulance, High) Emergency Priority (Fire Truck, High)

or a relation:

Approaching (Ambulance, Intersection1) Approaching (FireTruck, Intersection1)

The current KB is sufficient for independent emergency vehicles, but it is not sufficient for conflicting simultaneous emergencies at the same location.

Case Study 3 — Agricultural Expert Reasoning System

The agricultural knowledge base contains rules connecting soil dryness, irrigation, drip irrigation, and crop risk.

Predicates

Let:

- S: Soil Dry
- I: Irrigation Needed
- W: Crop Wheat
- D: Apply Drip Method
- R: Crop At Risk

The rules are:

P1: $S \rightarrow I$ P2: $I \wedge W \rightarrow D$ P3: $\neg D \wedge W \rightarrow R$

Facts:

S W

(a) Convert all sentences into CNF

P1

Original:

$S \rightarrow I$

Eliminate implication:

$\neg S \vee I$

Therefore:

P1: $\neg S \vee I$

P2

Original:

$I \wedge W \rightarrow D$

Eliminate implication:

$\neg(I \wedge W) \vee D$

Apply De Morgan's law:

$(\neg I \vee \neg W) \vee D$

Therefore:

P2: $\neg I \vee \neg W \vee D$

P3

Original:

$\neg D \wedge W \rightarrow R$

Eliminate implication:

$\neg(\neg D \wedge W) \vee R$

Using De Morgan's law:

$(D \vee \neg W) \vee R$

Therefore:

P3: $D \vee \neg W \vee R$

Facts

The facts are:

S=True
and:
W=True
In CNF:
S W

Complete CNF Knowledge Base

$\{\neg S \vee I, \neg I \vee \neg W \vee D, D \vee \neg W \vee R, S, W\}$

(b) Resolution Refutation — Prove 'Apply Drip Method'

KB in CNF:

C1: $\neg \text{Soil Dry} \vee \text{Irrigation Needed}$
C2: $\neg \text{Irrigation Needed} \vee \neg \text{Crop Wheat} \vee \text{Apply Drip Method}$
C3: $\text{Apply Drip Method} \vee \neg \text{Crop Wheat} \vee \text{Crop At Risk}$
C4: Soil Dry
C5: Crop Wheat
C6 (negated goal): $\neg \text{Apply Drip Method}$

Resolution proof tree:

Step 1: C1 + C4 $\rightarrow$ resolve on Soil Dry $\Rightarrow$ Irrigation Needed
Step 2: C2 + C1 $\rightarrow$ resolve on Irrigation Needed $\Rightarrow \neg \text{Crop Wheat} \vee \text{Apply Drip Method}$
Step 3: C3 + C5 $\rightarrow$ resolve on Crop Wheat $\Rightarrow$ Apply Drip Method
Step 4: C9 + C6 $\rightarrow$ resolve on Apply Drip Method

Therefore, Apply Drip Method is logically entailed by KB

(c) Remove Soil Dry from the Knowledge Base

Originally we have:

S

and:

W

Suppose S is removed.

The remaining fact is:

W

The rules remain:

$S \rightarrow I \quad I \wedge W \rightarrow D \quad \neg D \wedge W \rightarrow R$

Step 1

Can we derive IrrigationNeeded?

Rule:

$S \rightarrow I$

But S is no longer available.

Therefore:

I cannot be derived

Step 2

Can we derive ApplyDripMethod?

Rule:

$I \wedge W \rightarrow D$

We have:

W

but we do not have:

I

Therefore:

D cannot be derived

Step 3

Can we derive Crop At Risk?

Rule:

$\neg D \wedge W \rightarrow R$

We have:

W

But we do not have:

$\neg D$

The absence of D does not logically mean:

$\neg D$

This is an important principle of classical logic:

Failure to prove something is not the same as proving its negation.

Therefore:

CropAtRisk cannot be derived

Case Study 4 — Wumpus World Knowledge Agent

The agent perceives:

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

The given rules describe possible Wumpus/pit locations, gold, and safe cells.

Premise (fact)	Rule Applied	Conclusion Derived
Stench[1,2]	Stench[1,2] $\rightarrow$ Wumpus Adjacent[1,2]	Wumpus Adjacent[1,2] = True
Breeze[1,1]	Breeze[1,1] $\rightarrow$ Pit Adjacent[1,1]	Pit Adjacent[1,1] = True
Glitter[2,2]	Glitter[x,y] $\rightarrow$ Gold[x,y] (x=2,y=2)	Gold[2,2] = True

All derived conclusions:

- Wumpus Adjacent[1,2] — the Wumpus occupies one of the cells adjacent to [1,2]: {[1,1], [1,3], [2,2]}
- Pit Adjacent[1,1] — a Pit occupies one of the cells adjacent to [1,1]: {[1,2], [2,1]}
- Gold[2,2] — the gold is confirmed to be at cell [2,2]

(b) Forward Chaining to Identify Wumpus and Pit Locations

Forward chaining begins with the percepts and applies rules to derive additional facts.

Step 1

Initial fact:

Stench[1,2]

Apply Rule 1:

Stench[1,2] $\rightarrow$ Wumpus Adjacent[1,2]

Therefore:

Wumpus Adjacent[1,2]

Possible adjacent cells:

[1,1],[1,3],[2,2]

Step 2

Initial fact:

Breeze[1,1]

Apply Rule 2:

Breeze[1,1] $\rightarrow$ Pit Adjacent[1,1]

Therefore:

Pit Adjacent[1,1]

Possible adjacent cells:

[1,2],[2,1]

Step 3

Initial fact:

Glitter[2,2]

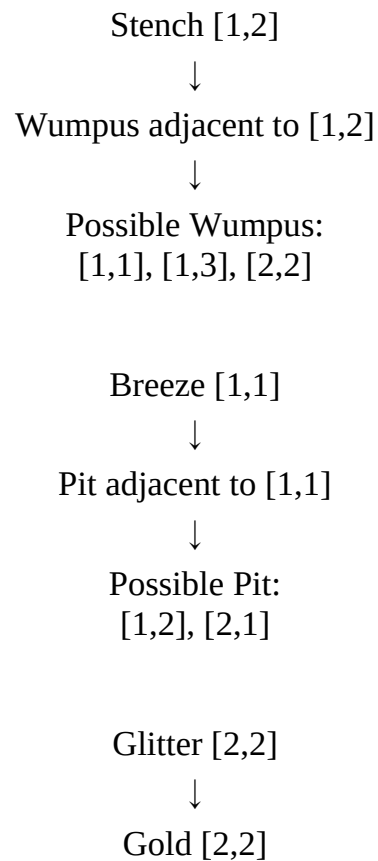
Apply Rule 3:

Glitter[2,2] $\rightarrow$ Gold[2,2]

Therefore:

Gold[2,2]

Forward Chaining:



Therefore, the logically justified

Wumpus {[1,1],[1,3],[2,2]}

(c) Is the Path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ Safe?

The proposed path is:

$[1,1] \rightarrow [1,2] \rightarrow [2,2]$

We must evaluate each cell.

Cell	Knowledge Available	Safe[x,y] Provable?	Verdict
[1,1]	Starting cell; agent has already survived being there	Yes — $\neg$ Wumpus and $\neg$ Pit both hold by initial assumption	Safe
[1,2]	Wumpus candidate from step (b) is $\{[1,3],[2,2]\}$, so $\neg$ Wumpus[1,2] holds; but [1,2] IS one of the two Pit candidates from PitAdjacent[1,1]	No — $\neg$ Pit[1,2] cannot be proven (Pit could be here)	Not confirmed safe — risk of Pit
[2,2]	This cell is one of the two remaining Wumpus candidates $\{[1,3],[2,2]\}$; Gold is confirmed here	No — $\neg$ Wumpus[2,2] cannot be proven (Wumpus could be here)	Not confirmed safe — risk of Wumpus

Therefore, $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is NOT proven safe by the current KB.

The agent needs additional percepts or knowledge, such as explicit evidence that particular cells contain no Wumpus and no pit, before declaring the route safe.